Milestone 04

PDF edition of preserved repository document | docs/milestone_04.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Research-core milestone: controlled numerical-claim study

IMPLEMENTED

An additional numerical-claim environment and backend, preserving all previous phases. Identical-capacity answer-only/structured baselines and numerical-claim Mindscape B/C; non-singleton constrained/unconstrained execution, scalar-feedback experience collection and positive SQLite replay, model-visible/oracle separation, bounded tagged simulation, optional offline pretrained encoder/head backend, four-condition study orchestration, component interventions, robustness analysis, statistics, fairness audit and SVG plots.

TESTED

All prior tests retained. Added numerical-claim oracle cases, wrong-value non-correction, 100 candidates, target-mutation feature isolation, equal capacity, three-way evidence records, no Regime C target access, non-mutating dream rollouts, generic learned-transition adapter, action modes, open-loop evidence removal, serialization/distractor invariance, all-condition training/reload/shared evaluation, optional offline backend contract (mocked), final-attempt completion counters, statistical utilities and invalid configuration rejection.

VERIFIED

90 trained checkpoints; 204 common-runner evaluations: 168 primary and 36 component/robustness evaluations. Complete primary matrix: four conditions × seven budgets × three seeds × two splits. All predictions/configs/checkpoints/failures retained locally. 17 artifact-based integrity checks passed; primary candidate count is 101, constrained count is 100. No arithmetic correction channel or model-visible inference verifier judgment is present.

MODEL(S) USED

Local task-specific NumPy tanh MLPs. The prior model remains intact. Optional offline LocalHFEncoder/FrozenPretrainedBackend supports pretrained text features plus a trainable head and explicit download opt-in; no actual pretrained weights were trained/evaluated in this study because PyTorch/Transformers are absent. Its offline interface is tested with a mock encoder. No model download, paid API or foundation-model training occurred.

MODEL SIZES

All four primary conditions: 52 inputs, 64 tanh units, eight ten-class output heads plus sign; 8,722 trainable parameters each. Policy uses first two heads for proposed unit/carry and other heads as zero dummy labels. Exact capacity is matched, but usable label semantics/task difficulty differ. Same model family, initialization policy and optimizer.

TRAINING REGIMES

A: final-answer supervision. Structured baseline: static structured operands -> answer. B: local numerical claims supervised by correct training transitions. C: bounded exploratory proposals -> scalar accepted/rejected feedback -> replay of accepted own labels, with no reference answers/trajectories read. Fixed 600 Adam steps, batch 64, learning rate .003; validation is diagnostic only, not checkpoint selection. Online/no-replay intervention uses one gradient update per accepted proposal and is not a matched-compute memory ablation.

DATASET SIZES

Research v1 (new frozen dataset): train 1000, validation 100, IID test 200, OOD test 300. Nested budgets: 10,25,50,100,250,500,1000. Train structures: 1x1/2x1/2x2; OOD: 3x2/3x3/4x2. Canonical operand pairs are disjoint, including swapped pairs. Finite 1x1 family exhausts in train; actual per-family/carry distributions remain in difficulty reports.

SEEDS

Dataset seed 4242; training/experience seeds 0,1,2 for every primary budget/condition. All interventions use three seeds at budget 50. All conditions share exact subset IDs.

ACTUAL RESULTS

Observed performance is low. None of the requested accuracy/OOD/groundedness/goal or data-efficiency targets is met. The largest primary IID score of any single run is 8%.

The prior 82% exploratory OOD result is not reproduced under numerical-claim controls.

No architecture-only superiority or 2×/5× data-efficiency conclusion is established.

IID RESULTS

Mean ± sample standard deviation across three seeds, percentages:

Budget	Baseline A	Baseline B	Mindscape B	Mindscape C
10	0.000 ± 0.000	0.000 ± 0.000	0.667 ± 0.289	0.167 ± 0.289
25	0.333 ± 0.289	0.000 ± 0.000	2.333 ± 0.764	0.667 ± 0.289
50	0.500 ± 0.000	0.500 ± 0.000	3.333 ± 0.577	1.333 ± 0.577
100	1.000 ± 0.000	0.833 ± 0.289	4.000 ± 0.000	2.333 ± 0.764
250	0.333 ± 0.289	0.500 ± 0.000	4.333 ± 0.577	4.000 ± 1.000
500	0.333 ± 0.289	0.333 ± 0.289	4.000 ± 0.866	5.500 ± 2.291
1000	0.167 ± 0.289	0.333 ± 0.289	6.333 ± 1.443	6.000 ± 1.732

At budget 1000, Mindscape B groundedness/goal success = 6.167% ± 1.155%; C = 5.667% ± 1.443%. Answer baselines have no supporting trajectory: groundedness and goal success are 0%, supplied-trajectory validity is null. For policy conditions, full-trajectory validity equals grounded/goal rates in these runs. Some wrong claims cancel to a correct final answer, so exact accuracy can exceed groundedness.

Unsupported rate is the complement of groundedness. Every metric remains in aggregate.json.

OOD RESULTS

All primary conditions/budgets/seeds have 0% exact OOD accuracy and 0% grounded/goal success on the observed 300-example OOD set. This does not mean a universal probability of zero; individual-run Wilson intervals are stored in statistics.json. Per-structure and carry counts are reported in difficulty_failures.json. No held-out test tuning occurred.

DATA-EFFICIENCY RESULTS

N*(.80), N*(.90), N*(.95) and both baseline-to-Mindscape DERs are not_reached on IID and OOD. No interpolation or fabricated crossing. Ablation data-efficiency changes are not estimated from single-budget interventions.

ABLATION RESULTS

Mean IID accuracy changes at budget 50, percentage points (three seeds):

Intervention	Reference	ΔAccuracy	ΔGroundedness	ΔGoal success
constrained	trajectory	0.000	0.000	0.000
no_dream	trajectory	-0.333	-0.167	-0.167
no_environment_interaction	trajectory	0.000	-2.833	-2.833
no_episodic_replay	experiential	-1.333	-0.500	-0.500
no_state	trajectory	-3.333	-2.833	-2.833
serialization_distractor	trajectory	0.000	0.000	0.000

All OOD intervention changes are 0 percentage points. No-interaction preserves model-computed answers while removing actual evidence; the groundedness drop follows the metric definition, not proof of an arithmetic advantage. Serialization/distractor invariance is a deterministic feature-adapter property, not semantic distractor reasoning. Constant relation representation and mandatory external goal evaluator are not meaningful isolated ablations in this domain. B versus C varies trajectory supervision by construction. Reasons are recorded in ablations.json. The online/no-replay contrast also changes optimizer and update count; it cannot isolate memory causally at equal compute.

EXPERIENTIAL-LEARNING STATUS

Implemented limited batch off-policy self-generated feedback learning. At budget 1000, accepted replay labels are 738/874/1074 at seeds 0/1/2, versus 4506 supervised transitions per B run. Query budget: up to 64 per problem; exact counts are retained. No complete expert trace is delivered to C. Exploration is collected with initial weights, then positive feedback is replayed; no claim of online adaptive exploration or autonomous concept discovery. Sparse feedback and limited capacity are potential limitations, not

experimentally isolated explanations for poor performance.

DREAM-ENGINE STATUS

Runnable generic bounded branching with explicitly hypothetical states and deterministic or learned-callable TransitionModel adapters. Primary claim dreams use proposed-write dynamics, depth 2/branching 3, and learned confidence scoring without a numerical oracle. Goal-distance scoring is demonstrated separately in examples/dream_goal_demo.json: real state remains 0, goal is 3, selected first action is +1. A separately trained transition world model is not implemented. Simulated values never become observed evidence.

HISTORICAL-REFERENCE STATUS

Primary-source GPT-4/Gemini report records and exact prompting caveats are documented in docs/historical_references.md. GSM8K word problems are not comparable to our explicit integer multiplication task; historical checkpoint IDs are undisclosed and training budgets are not controlled domain sample counts. No victory claim or historical DER.

FAIRNESS CHECKS

17 computed checks passed: dataset integrity/disjointness, training IDs, equal parameter count, dataset hashes, balanced matrix, shared optimizer configuration, no target reads in C, no inference arithmetic-oracle calls, target-mutation isolation, >1 candidates, no inference correctness feedback, own-proposal actual values, hypothetical tags and train-only memory. This removes major earlier confounds, but hand-specified decomposition, different label/query counts, compute and trace accessibility remain explicit differences. No claim that architecture alone has been isolated. Exploratory paired McNemar tests at budget 1000, seed 0, after Holm correction: B vs A IID p=0.0001831, C vs A IID p=0.0390625; all other comparisons p=1. These within-seed tests concern small absolute IID differences versus a weak baseline and do not establish practical success or OOD benefit.

TRAINING TIME

32.965690 seconds across 90 trained primary/intervention models; actual per-run metadata retained.

INFERENCE TIME

114.325489 seconds across 204 evaluations; each covers 200 IID or 300 OOD examples. Includes model execution and evaluator assessment.

TEST COUNT

86 passed on Python 3.12.14 / NumPy 2.3.5; original tests preserved. Real optional PyTorch/Transformers integration, pytest/Ruff/MyPy and hosted CI were not run.

KNOWN LIMITATIONS

All primary OOD runs fail; IID success is far below targets. Task MLPs are small and not pretrained; optional pretrained interface was only mock-tested. The procedure/control schedule is hand-authored and narrows the scientific question to decomposed numerical learning. Goal/relation are fixed domain signals, not variable-goal or graph reasoning. Label/query budget and inference compute differ; replay uses only accepted labels and does not optimize rejected-action likelihood. Baselines cannot meet trace-groundedness without a trace emitter. Scalar C feedback uses a correctness oracle at training only, an intentional experimental condition. No final thesis conclusions or measured 2× DER.

A bookkeeping edge case undercounted completed C episodes in two original metadata files (1000/seed0: 23 vs stored 24; 1000/seed2: 76 vs stored 77). The audit retains both values; completion reporting here uses stored outcomes. Future code counts final-attempt completion correctly, with a regression test. Labels, weights and evaluation metrics were unaffected. Original experiment files were not replaced. Analysis-only chart drafts are retained; committed figures use corrected cost labels and uncertainty bars.
Local results occupy approximately 1.0 GiB, including replay/prediction detail. Checkpoints and full predictions stay local under ignored results/; measured summary snapshots are committed. Timing excludes dataset preparation and artifact I/O for evaluation; training includes experience collection/replay and fitting. RAM size was unavailable. PNG rendering is absent, so figures are standalone SVG vectors. Claims adapter is currently domain-specific.

GIT STATUS

Branch main; earlier code/history/results preserved. Final clean status checked after commit.

LATEST COMMIT

Protocol/run revision: df98891. Analysis/backend/diagnostics: 571147b. Final report and measured snapshots are committed separately; latest hash is reported in chat.

Diagnostic naming caveat: original C metadata used the generic key validation_metrics for training-replay diagnostics, not held-out validation. diagnostic_context.json sidecars identify the actual source without replacing original files; future metadata separates replay_metrics from validation_metrics. No C validation trace labels were used and no reported held-out benchmark metric depends on these training diagnostics.

REPOSITORY TREE

```
src/mindscape/environments/multiplication/claims.py
src/mindscape/data/claims_backend.py
src/mindscape/models/{study,study_encoding,pretrained}.py
src/mindscape/training/claims.py
src/mindscape/reasoning/simulator.py
configs/experiments/{research_dataset,research_study}.toml
datasets/generated/research_v1/{manifest.json,*.jsonl}
scripts/{train_research,evaluate_research,run_research_study}.py
scripts/{analyze_research_study,audit_research_study,run_dream_demo}.py
docs/{research_protocol,historical_references,milestone_04}.md
experiments/research_claims_v1/{aggregate,data_efficiency,ablations,statistics,...}.json
experiments/research_claims_v1/plots/*.svg
results/research_claims_v1/<condition_budget_seed>/{config.yaml,metadata.json,checkpoint,t
raining.jsonl}
results/research_claims_v1/evaluations/<id>/{config.yaml,metrics.json,predictions.jsonl,fa
ilures.json,summary.md,plots/}
examples/dream_goal_demo.json
tests/test_research.py
```